

Symbolic Regression of the cDFT–WSD Surface-Tension Residual for CO₂-Rich Mixtures

A Comparison of a Physics-Informed (V1) and a Composition-Only (V2) Feature Set

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Abstract

We learn a closed-form correction to the uncorrected White-Bear / weighted solvation-density (WSD) surface tension, so that it reproduces the full classical density-functional-theory (cDFT) prediction for CO₂-rich mixtures. The learning target is the dimensionless *relative residual* $\varepsilon = (\gamma_{\text{cDFT}} - \gamma_{\text{base}}) / \gamma_{\text{base}}$. Two feature-engineering strategies are compared under an identical train/validation/test protocol ($n = 23\,040$, seed 5002034): **V1**, a twelve-feature set that augments the composition aggregates with physics-informed reduced state variables (reduced temperature and pressure, saturation proximity, and the Parachor density gap); and **V2**, a parsimonious five-feature composition-only set. For each strategy we report the full Pareto frontier, converting the symbolic-regression loss (mean squared error) to RMSE and computing the per-complexity *score* used for model selection. Generalisation is assessed by 5-fold cross-validation using the same stratified protocol as the GPR / SVGP residual surrogates. The selected physics-informed V1 model is a compact rational form (complexity 12) with a test $R^2 = 0.941$ on the residual and near-identical train/validation/test scores; it reconstructs γ_{cDFT} to within $0.074\,00\,\text{mN m}^{-1}$ MAE. Cross-validation of the V1 feature set gives $R^2 = 0.961 \pm 0.005$, confirming its robustness. The interpretable V2 law, $\varepsilon \approx -2.555\,x_{\text{imp}}$, captures the leading-order behaviour at complexity 4 with a test $R^2 = 0.618$.

1 Problem Statement and Target Definition

The baseline model is the uncorrected WSD surface tension, $\gamma_{\text{base}} \equiv \gamma_{\text{wsd}}^{\text{UC}}$, and the reference is the full cDFT prediction reconstructed as

$$\gamma_{\text{cDFT}} = \gamma_{\text{wsd}}^{\text{UC}} + (\gamma_{\text{cDFT}} - \gamma_{\text{wsd}})_{\text{uncorrected}}. \quad (1)$$

Rather than regressing γ_{cDFT} directly, we regress the *dimensionless relative residual*

$$\boxed{\varepsilon = \frac{\gamma_{\text{cDFT}} - \gamma_{\text{base}}}{\gamma_{\text{base}}}} \quad (2)$$

so that the baseline carries the dominant signal and symbolic regression need only describe the (small) physical correction. Predictions are mapped back to dimensional surface tension through

$$\widehat{\gamma_{\text{cDFT}}} = \gamma_{\text{base}} (1 + \widehat{\varepsilon}), \quad (3)$$

which makes the reported γ_{cDFT} metrics directly comparable to the GPR/SVGP surrogates trained on the same data.

2 Data and Protocol

All cases are drawn from `CombinedDatasetSEC.A4.csv`. After dropping rows with missing or non-finite features the dataset contains $n = 23\,040$ points, split 70/15/15 with a fixed `MersenneTwister` seed of 5002034:

	Train	Validation	Test
Rows	16 130	3456	3457

The split, seed, and target (2) are identical across both feature sets, so any difference in performance is attributable solely to the *features* and the search budget.

3 Feature Engineering

The impurity is defined as every component other than CO_2 . Liquid-, vapour-, and feed-phase impurity sums are formed as $x_{\text{imp}} = \sum_{i \neq \text{CO}_2} x_i$, $y_{\text{imp}} = \sum_{i \neq \text{CO}_2} y_i$, and $z_{\text{imp}} = \sum_{i \neq \text{CO}_2} z_i$, together with the phase-partitioning difference $\Delta_{\text{imp}} = x_{\text{imp}} - y_{\text{imp}}$ and the impurity partition coefficient $\ell_K = \ln[(y_{\text{imp}} + \epsilon)/(x_{\text{imp}} + \epsilon)]$ with $\epsilon = 10^{-12}$. These five *composition aggregates* are common to both versions.

V1 – physics-informed (12 features). The composition aggregates are augmented with reduced state variables referenced to pure CO_2 , motivated by the strong dependence of surface-tension residuals on critical-point proximity and on the density gap (Macleod–Sugden / Parachor variable):

$$\begin{aligned}
 T_r &= T/T_c, & (1 - T_r) &= 1 - T_r, & P_r &= P/P_c, \\
 \Pi &= P/P_{\text{sat}}^0, & \Delta\rho &= \rho_l - \rho_v, & \Delta\rho^* &= \frac{\rho_l - \rho_v}{\rho_l^0 - \rho_v^0}, \\
 \gamma_r &= \gamma_{\text{wsd}}/\gamma_{\text{CO}_2}^0.
 \end{aligned}$$

V2 – composition-only (5 features). The five composition aggregates $\{z_{\text{imp}}, x_{\text{imp}}, y_{\text{imp}}, \Delta_{\text{imp}}, \ell_K\}$ only; all state and density-gap variables are deliberately excluded to obtain a maximally interpretable, composition-driven law.

Table 1 summarises the symbols used in the equations below.

Table 1: Notation for the regressed features.

Symbol	Feature	Symbol	Feature
x_{imp}	<code>x_total_impurity</code>	T_r	<code>Tr</code>
y_{imp}	<code>y_total_impurity</code>	$(1 - T_r)$	<code>oneMinusTr</code>
z_{imp}	<code>z_total_impurity</code>	P_r	<code>Pr</code>
Δ_{imp}	<code>delta_total_impurity</code>	Π	<code>P_over_Psat0</code>
ℓ_K	<code>logK_total_impurity</code>	$\Delta\rho$	<code>delta_rho</code>
γ_r	<code>gamma_base_ratio</code>	$\Delta\rho^*$	<code>delta_rho_norm</code>

4 Symbolic Regression Configuration

Both runs use `SymbolicRegression.jl` with multithreaded `equation_search` over 200.0 iterations and protected operators. The two configurations differ only in operator richness, population count and the complexity ceiling (Table 2).

Table 2: Search configuration for the two feature sets.

	V1 (physics-informed)	V2 (composition)
Binary operators	<code>+</code> <code>-</code> <code>×</code> , <code>safepow</code> , <code>safe_div</code>	<code>+</code> <code>-</code> <code>×</code> , <code>safepow</code> , <code>safe_div</code>
Unary operators	<code>abs</code> , <code>safe_sqrt</code> , <code>safe_log</code>	<code>abs</code>
Populations	120	80
Max. size	24	16
Parsimony	0.003	0.003
Batching	1000	1000

4.1 Selection criterion: from loss to RMSE and score

The search minimises the mean squared error of ε , reported as the *loss* L . Because L is a squared quantity we report the root mean squared error

$$\text{RMSE} = \sqrt{L}, \quad (4)$$

which is on the same scale as ε itself. To rank Pareto-optimal candidates we use the standard per-complexity *score*, i.e. the rate at which the loss decreases as one unit of complexity c is added:

$$\text{score}_i = - \frac{\ln L_i - \ln L_{i-1}}{c_i - c_{i-1}}. \quad (5)$$

A large score marks an equation whose added complexity is well justified by the accuracy it buys; once the score collapses towards zero, further complexity is essentially “free-riding” on overfitting. The final model is chosen as the best trade-off between low RMSE and a still-meaningful score, rather than the lowest-loss (largest) expression.

5 Pareto Frontiers

5.1 V1 – physics-informed

Table 3 lists the complete Pareto frontier. The score (5) is large at low complexity (a constant and the $\ln(\Delta\rho^*)$ family) and falls steeply beyond complexity 7. Around complexity 10 the frontier transitions to a *rational* form, $\ln(\Delta\rho^*)$ divided by a pressure/composition denominator, which carries the accuracy-dominant tail. We select the **complexity-12** member (★): it is the simplest equation on this rational branch and sits at the “knee” of the frontier, capturing $R^2 = 0.941$ with less than half the complexity of the most accurate (complexity-24) expression, whose extra terms buy only a further ~ 1.5 points of R^2 (see Sec. 6).

5.2 V2 – composition-only

The V2 frontier (Table 4) is dominated by a single sharp score at complexity 2 and a strong secondary gain at complexity 4. Beyond complexity 4 the RMSE plateaus near 0.027: the additional terms reduce the loss by less than 15.00% while quadrupling the complexity, so the selected model is the complexity-4 linear law (★).

6 Selected Models

V1 (physics-informed), complexity 12.

$$\hat{\varepsilon}_{\text{V1}} = \frac{\ln(\Delta\rho^*)}{\sqrt{P_r} + (\gamma_r - \Delta\rho^*)}. \quad (6)$$

Table 3: V1 Pareto frontier. L is the SR loss (MSE of ε); RMSE and MAE are evaluated on the training set; score from Eq. (5). ★ marks the selected equation.

c	RMSE	MAE	L	score	Equation
1	0.0849	0.0740	$7.203 \cdot 10^{-3}$	—	$\Delta\rho$
2	0.0431	0.0342	$1.855 \cdot 10^{-3}$	1.357	-0.0526
3	0.0239	0.0178	$5.698 \cdot 10^{-4}$	1.180	$\ln(\Delta\rho^*)$
4	0.0170	0.0130	$2.890 \cdot 10^{-4}$	0.679	$\Delta\rho^* - 1.0183$
5	0.0141	0.0085	$1.983 \cdot 10^{-4}$	0.377	$\ln(\Delta\rho^*) - x_{\text{imp}}$
6	0.0127	0.0084	$1.624 \cdot 10^{-4}$	0.199	$\Delta\rho^* - (x_{\text{imp}} + 0.9979)$
7	0.0124	0.0083	$1.526 \cdot 10^{-4}$	0.062	$(\ln(\Delta\rho^*) - x_{\text{imp}}) \Delta\rho^*$
9	0.0123	0.0082	$1.510 \cdot 10^{-4}$	0.005	$(\Delta\rho^* - \Delta\rho) \ln(\Delta\rho^* - x_{\text{imp}})$
10	0.0111	0.0075	$1.235 \cdot 10^{-4}$	0.202	$\ln(\Delta\rho^*) / \sqrt{P_r} \gamma_r$
12	0.0104	0.0071	$1.082 \cdot 10^{-4}$	0.066	$\ln(\Delta\rho^*) / [\sqrt{P_r} + (\gamma_r - \Delta\rho^*)]$ ★
15	0.0097	0.0067	$9.497 \cdot 10^{-5}$	0.044	$P_r^{(1-T_r)-y_{\text{imp}}} T_r (\ln(\Delta\rho^*) - x_{\text{imp}})$
16	0.0097	0.0066	$9.410 \cdot 10^{-5}$	0.009	$P_r^{-0.502 y_{\text{imp}}} T_r (\ln(\Delta\rho^*) - x_{\text{imp}})$
17	0.0097	0.0065	$9.356 \cdot 10^{-5}$	0.006	$T_r \ln(\Delta\rho^* - x_{\text{imp}}) \sqrt{P_r^{\Delta_{\text{imp}}} - z_{\text{imp}}}$
18	0.0094	0.0065	$8.893 \cdot 10^{-5}$	0.051	$P_r^{(1.491-\gamma_r)\Delta_{\text{imp}}} T_r (\ln(\Delta\rho^*) - x_{\text{imp}})$
19	0.0093	0.0061	$8.569 \cdot 10^{-5}$	0.037	$\ln(\Delta\rho^*) / [P_r^{0.266} + \ell_K ((\gamma_r - \Delta\rho^*) - z_{\text{imp}})]$
21	0.0090	0.0060	$8.047 \cdot 10^{-5}$	0.031	$\ln(\Delta\rho^*) / [P_r^{0.279} + \ell_K (\gamma_r (\gamma_r - \Delta\rho^*) - z_{\text{imp}})]$
23	0.0089	0.0059	$7.878 \cdot 10^{-5}$	0.011	$\ln(\Delta\rho^*) / [(\ell_K - \Delta\rho^*)(\gamma_r - \Delta\rho^*) + P_r^{0.352} - 2z_{\text{imp}}]$
24	0.0088	0.0058	$7.742 \cdot 10^{-5}$	0.017	$\ln(\Delta\rho^*) / [P_r^{0.279} + \ell_K \gamma_r ((\gamma_r - \Delta\rho^*) - z_{\text{imp}})] - 0.0021$

The correction takes a compact *rational* form: the numerator is the logarithm of the normalised density gap $\Delta\rho^*$ (Parachor variable), while the denominator combines a square-root reduced-pressure term $\sqrt{P_r}$ with the baseline- γ /density-gap difference $(\gamma_r - \Delta\rho^*)$. This links the residual to interfacial density contrast (numerator) and to pressure / density state (denominator) with only four variables and no fitted constants. It is the simplest member of the rational branch of the frontier and its train, validation and test scores agree to three decimals ($R^2 = 0.9416/0.9412/0.9408$), evidence that the form generalises rather than memorises.

V2 (composition-only), complexity 4.

$$\hat{\varepsilon}_{\text{V2}} = -2.5549 x_{\text{imp}}. \quad (7)$$

To leading order the residual is simply *proportional to the liquid-phase impurity fraction*: every percent of impurity in the liquid lowers γ_{cDFT} relative to the WSD baseline by $\approx 2.550\%$.

Table 5 collects both selected equations side by side.

7 Performance

Table 6 reports the held-out performance of the two selected models, both in the dimensionless residual space (ε) and after reconstruction to surface tension via Eq. (3).

8 Cross-Validation

To assess generalisation independently of the single train/test split, each feature set is evaluated by **5-fold cross-validation** using the same protocol as the GPR / SVGP residual surrogates: a stratified k -fold over T-P decile bins (shuffled, seed 4555525). Because symbolic regression refits the entire search each fold, the CV measures the generalisation of the *feature set and search* – analogous to refitting the GPR kernel per fold – rather than of one fixed equation; within each

Table 4: V2 Pareto frontier (columns as in Table 3).

c	RMSE	MAE	L	score	Equation
1	0.0909	0.0730	$8.271 \cdot 10^{-3}$	—	x_{imp}
2	0.0431	0.0342	$1.855 \cdot 10^{-3}$	1.495	-0.0526
4	0.0270	0.0180	$7.291 \cdot 10^{-4}$	0.467	$-2.5550 x_{\text{imp}} \star$
5	0.0270	0.0180	$7.291 \cdot 10^{-4}$	0.000	$x_{\text{imp}}/(-0.3914)$
6	0.0270	0.0179	$7.277 \cdot 10^{-4}$	0.002	$x_{\text{imp}}(z_{\text{imp}} - 2.5972)$
7	0.0269	0.0178	$7.222 \cdot 10^{-4}$	0.008	$x_{\text{imp}}/(-0.3492 - z_{\text{imp}})$
8	0.0268	0.0177	$7.161 \cdot 10^{-4}$	0.008	$x_{\text{imp}}(-2.3588 - \Delta_{\text{imp}}^2)$
9	0.0267	0.0177	$7.151 \cdot 10^{-4}$	0.001	$-2.3785 x_{\text{imp}} + 0.0158 \Delta_{\text{imp}}$
10	0.0265	0.0175	$7.007 \cdot 10^{-4}$	0.020	$-3.3478 y_{\text{imp}}^{y_{\text{imp}}} x_{\text{imp}}$
11	0.0263	0.0174	$6.925 \cdot 10^{-4}$	0.012	$(-2.3320 - -0.3671 - \Delta_{\text{imp}}) x_{\text{imp}}$
12	0.0262	0.0173	$6.883 \cdot 10^{-4}$	0.006	$(y_{\text{imp}} - z_{\text{imp}})^{y_{\text{imp}}} (-3.5244 x_{\text{imp}})$
13	0.0255	0.0168	$6.505 \cdot 10^{-4}$	0.057	$x_{\text{imp}}(-1.7925 - \ell_K \Delta_{\text{imp}} + 1.0744)$
15	0.0254	0.0166	$6.460 \cdot 10^{-4}$	0.003	$x_{\text{imp}}(-2.0208 - y_{\text{imp}} + \Delta_{\text{imp}} \ell_K + 0.6315)$
16	0.0253	0.0166	$6.417 \cdot 10^{-4}$	0.007	$x_{\text{imp}}(-1.9235 - -0.5519 - \ell_K(\Delta_{\text{imp}} + 0.2337))$

Table 5: Selected equations for both versions.

Version	c	Test R^2	Selected $\hat{\varepsilon}$
V1	12	0.941	$\ln(\Delta \rho^*) / [\sqrt{P_r} + (\gamma_r - \Delta \rho^*)]$
V2	4	0.618	$-2.5550 x_{\text{imp}}$

fold the equation is selected by an inner validation split (lowest validation RMSE). Scores are reported as mean $\pm 2\sigma$ across folds (Table 7).

The CV means track the single-split hold-out values in Table 6 closely, and the 2σ bands are narrow, confirming that the V1-over-V2 advantage is not an artefact of the chosen split. The V2 CV R^2 (0.661) exceeds its reported test value (0.618) only because the per-fold auto-selection settles on complexity-16 expressions rather than the hand-picked complexity-4 law of Eq. (7).

9 Diagnostic Plots of the Selected Equations

Figures 1 and 2 show the held-out behaviour of the two *selected* equations, Eqs. (6) and (7), in the dimensionless residual (ε) space. The parity plots (Fig. 1) overlay the train, validation and test splits against the $y = x$ line with per-split R^2 ; the residual plots (Fig. 2) show the prediction error versus the predicted value together with the $\pm$ RMSE bands on the test and validation splits.

10 Discussion

- **Adding physics pays off in residual space.** Moving from the composition-only V2 to the physics-informed V1 lifts the test R^2 on ε from 0.618 to 0.941 and roughly halves the reconstructed surface-tension error ($0.07400 \text{ mN m}^{-1}$ vs 0.1620 mN m^{-1} MAE). The reduced state and density-gap variables therefore encode information that pure composition cannot.
- **Parsimony at the knee of the frontier.** The score column (5) shows that the bulk of V1’s accuracy is already captured by complexity ≤ 7 , and the rational branch that begins near complexity 10 adds a further jump. We deliberately select the *complexity-12* equation rather than the most accurate complexity-24 expression: the former reaches $R^2 = 0.941$ with a four-variable, constant-free closed form, while doubling the complexity to 24 lifts

Table 6: Held-out performance of the selected V1 and V2 models. Residual metrics are on ε (dimensionless); γ_{cDFT} metrics are in mN m^{-1} .

	V1 (test)	V2 (test)	V1 (val)	V2 (val)
εR^2	0.941	0.618	0.9412	0.6207
ε RMSE	0.0102	0.0260	0.010 28	0.026 12
ε MAE	0.006 98	0.0174	0.007 110	0.017 49
$\gamma_{\text{cDFT}} R^2$	0.999	0.998	0.9995	0.9978
γ_{cDFT} RMSE	0.142	0.289	0.1424	0.2824
γ_{cDFT} MAE	0.0743	0.162	0.075 70	0.1606
Complexity	12	4	12	4
SR loss L	$1.08 \cdot 10^{-4}$	$7.29 \cdot 10^{-4}$		

Table 7: 5-fold cross-validation (mean $\pm 2\sigma$), $n = 23\,040$, 200 iterations/fold. Residual metrics are on ε ; γ_{cDFT} metrics are in mN m^{-1} . The last row is the per-fold selected complexity.

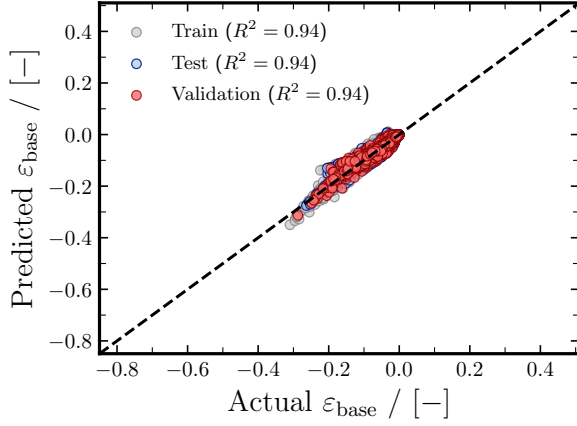
	V1 (physics-informed)	V2 (composition-only)
εR^2	0.961 ± 0.005	0.661 ± 0.014
ε RMSE	0.00842 ± 0.00058	0.02494 ± 0.00050
ε MAE	0.00568 ± 0.00063	0.01640 ± 0.00051
$\gamma_{\text{cDFT}} R^2$	0.99965 ± 0.00009	0.99831 ± 0.00030
γ_{cDFT} RMSE	0.113 ± 0.015	0.247 ± 0.022
γ_{cDFT} MAE	0.0603 ± 0.0061	0.1434 ± 0.0146
Complexity / fold	22–24	16

R^2 to only 0.956 – a 1.5-point gain that does not justify the loss of interpretability. Its train/validation/test scores agree to three decimals, confirming the simpler form is not under-fitting in a fragile way.

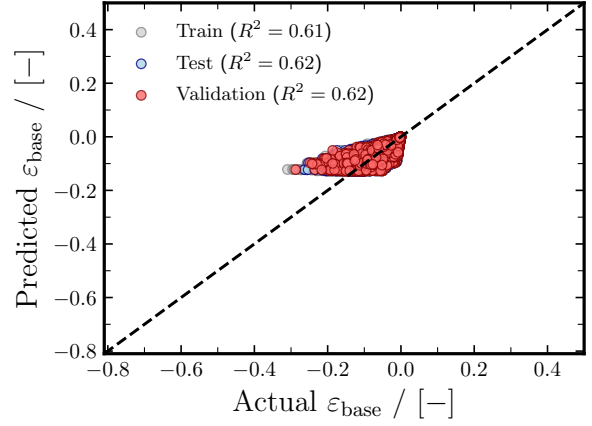
- **Cross-validation confirms the ranking is robust.** The 5-fold CV (Table 7) reproduces the hold-out gap with tight scatter: the V1 feature set reaches $\varepsilon R^2 = 0.961 \pm 0.005$ versus V2’s 0.661 ± 0.014 , and the small standard deviations show neither feature set over-fits a particular split. The CV auto-selects the lowest-validation-RMSE equation each fold (complexity 22–24 for V1, 16 for V2), so it measures the generalisation of the *feature set + search* and slightly exceeds the parsimonious selected equations (V1 complexity 12, V2 complexity 4) reported in Table 5.
- **The honest comparison is in ε space.** In absolute γ_{cDFT} both models exceed $R^2 = 0.997$, but this is largely inherited from the WSD baseline through Eq. (3). The residual-space metrics expose the true gap between the two feature sets.
- **V2 is the interpretable distillate.** Equation (7) gives a one-line physical statement – the cDFT correction is, to first order, linear in the liquid impurity fraction with slope -2.55 – which is valuable as a sanity check and a reduced-order model even though it is less accurate.

11 Conclusion

Two symbolic-regression strategies were applied to the cDFT–WSD surface-tension residual of CO_2 -rich mixtures under an identical protocol. The physics-informed V1 model, Eq. (6), a compact rational form, is the recommended predictive correction (test $R^2 = 0.941$ on ε , 5-fold-CV feature-set $R^2 = 0.961 \pm 0.005$, $0.074\,00\,\text{mN m}^{-1}$ MAE on γ_{cDFT}), chosen at the knee of the Pareto

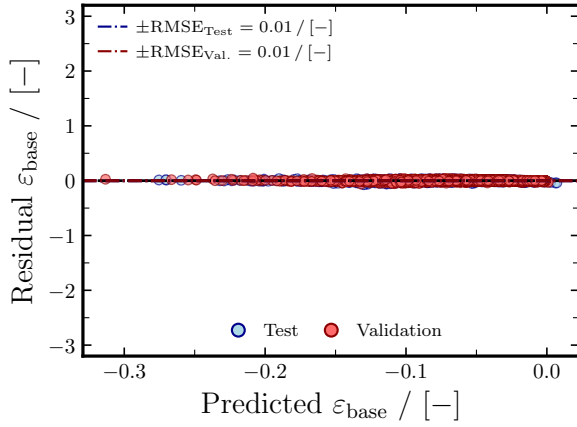


(a) V1 – physics-informed (complexity 12).

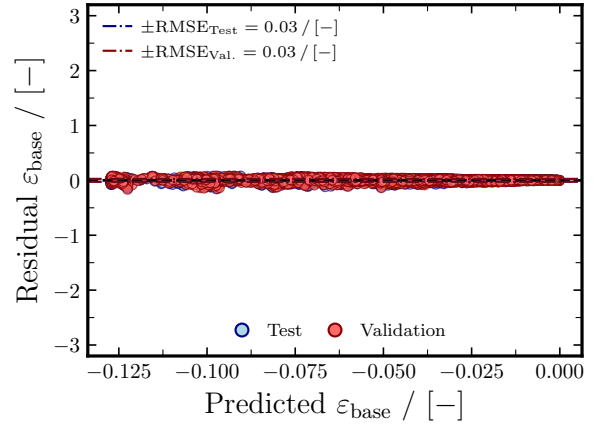


(b) V2 – composition-only (complexity 4).

Figure 1: Parity plots ($\hat{\varepsilon}$ vs ε) of the selected equations. The tighter clustering about $y = x$ for V1 reflects its higher test $R^2 = 0.941$ relative to V2’s $R^2 = 0.618$.



(a) V1 – physics-informed (complexity 12).



(b) V2 – composition-only (complexity 4).

Figure 2: Residual-vs-predicted plots of the selected equations, with the $\pm$ RMSE bands for the test and validation splits. The V1 band is roughly three times narrower than V2’s, consistent with the ε -RMSE values in Table 6.

frontier as the best balance of complexity, RMSE and per-complexity score. The composition-only V2 model, Eq. (7), is retained as a compact, interpretable leading-order law. Together they bracket the accuracy–interpretability trade-off of the surrogate correction.